

AIR+WATER

Smog

Smog is a type of air pollution that results from the combination of fog and smoke or other pollutants in the atmosphere.

It is characterized by a hazy, often yellowish appearance in the air and can dangerously reduce visibility.

Type of Smog:

- Photochemical Smog
- Industrial Smog

Types of Smog

Photochemical Smog:

This type of smog forms usually in urban areas with high levels of vehicular emissions.

It is a result of reactions between pollutants like nitrogen oxides (NO_x) and volatile organic compounds (VOCs) in the presence of sunlight. It contains ozone, which can be harmful to human health.

Industrial Smog:

It is also known as "black smog," this type results from burning coal or other fossil fuels.

It contains large amounts of particulate matter and sulphur dioxide.

Industrial smog is commonly

associated with areas where coal is the primary source of energy.

Composition:

- Nitrogen oxides (NO_x)
- Volatile organic compounds (VOCs)
- Particulate matter (PM)
- Sulphur dioxide (SO₂)

Hazardous Effects

1. Respiratory Problems:

Smog can exacerbate respiratory conditions like asthma, pharyngitis, bronchitis, and other lung diseases. It can cause irritation, coughing, and difficulty in breathing.

2. Cardiovascular Issues:

Prolonged exposure to smog is linked to heart problems, including heart attacks and arrhythmia.

3. Reduced Visibility:

Smog reduces visibility, increasing the risk of accidents on roads, air travel, and even maritime navigation.

4. Environmental Impact:

Smog damages crops, forests, and aquatic ecosystems. Ozone, a component of smog, harms plant growth and decreases crop yields.

5. Eye Irritation:

Smog can lead to redness, itching, and watering of the eyes.

6. Global Warming:

Some components of smog, like methane, are potent greenhouse gases that contribute to global Warming and climate change.

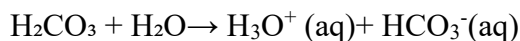
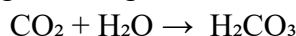
7. Aesthetic degradation:

Smog gives rise to a persistent haze, spoiling scenic views and impacting the overall aesthetic quality of an area.

Acid Rain

“Acid rain is a type of environmental pollution characterized by rainfall or precipitation that has a lower pH (less than 5.6) level than normal.”

Rain water does not have pH 7 due to slight mixing of CO₂ which forms a weak acid carbonic acid.



Reasons for Acid Rain

Emissions from Fossil Fuels:

The primary cause of acid rain is the release of sulphur dioxide (SO₂) and nitrogen oxides (NO_x) into the atmosphere, mainly from burning fossil fuels like coal and oil in power plants and vehicles.

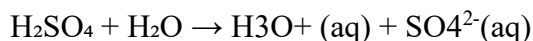
Industrial Processes:

Industrial activities, such as manufacturing and mining, also release these pollutants, contributing to acid rain.

Reactions Involved:

Formation of Sulphuric Acid (H₂SO₄):

Sulphur dioxide (SO₂) reacts with oxygen (O₂) and water (H₂O) to form sulphuric acid (H₂SO₄):



Formation of Nitric Acid (HNO₃):

Nitrogen oxides (NO_x) react with atmospheric moisture to form (HNO₃):



Impacts of Acid Rain:

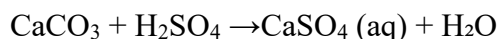
1. Environmental Damage:

Acid rain can harm forests, lakes, rivers, and aquatic ecosystems.

It damages plant foliage and aquatic life by altering the pH of soil and water, making it difficult for some organisms to survive.

2. Corrosion:

Acid rain accelerates the corrosion of buildings, statues, and infrastructure, causing significant damage to historical landmarks and urban structures.



3. Human Health:

Inhalation of pollutants associated with acid rain, such as fine particulate matter can have adverse health effects, including respiratory problems and cardiovascular issues.

4. Economic Costs:

Acid rain can lead to increased maintenance and repair costs for infrastructure, as well as reduced agricultural productivity due to soil degradation.

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7. Loss of Biodiversity:

Acid rain affects aquatic life by lowering pH levels in water bodies and formation of Aluminium based gelatinous compounds that clogs in gills, harming fish and amphibians.

It can also lead to the decline of sensitive plant and animal species in affected areas.

8. Global Implications:

While acid rain primarily affects local and regional areas, the pollutants responsible for its formation can travel long distances in the atmosphere, contributing to trans boundary pollution and affecting ecosystems in distant regions.

Conservation and Management strategies for protecting and preserving water resources**1. Smart use of water:**

A country should promote the efficient water use in homes, industries and agriculture through different technologies like drip irrigation system, low- flow fixture.

2. Wastewater treatment

Waste water treatment plants should be installed on small scale level and used water should be purified for reuse, it can save the freshwater.

3. Rainwater harvesting

For non-potable uses like irrigation and toilet flushing we can collect rainwater, this reduces the demand of freshwater.

4. Watershed's protection

We should restore and maintain natural watersheds for steady supply of water. This includes erosion control and reforestation in barren lands.

5. Water recycling

For irrigation and industrial processes, the reuse of wastewater should be promoted.

6. Legislation and regulation

We should enforce laws and regulations to prevent pollution control, water withdrawals, and ensure sustainable management of water reservoirs.

7. Public awareness

We should educate people about the importance of water, it's conservation and tell them the methods to reduce water pollution.

8. Desalination

The cities located near the oceans can use desalinated water as a source of freshwater. This can reduce burden on freshwater supplies.

9. Research and technology

Governments should invest in research for innovative technology for saving water and sustainable water management practices.

10. International cooperation

Countries should collaborate with neighbouring regions and other countries on trans boundary issues to ensure sustainable and equitable water use.

For More Information:

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